Introduction to Database Systems (CSD202)

Lab Assignment#1

Create database

```
mysql> create database DBMS2019;
Query OK, 1 row affected (0.03 sec)
```

Display list of databases

```
mysql> show databases
    -> ;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| adbms_19           |
| car_rental         |
| data               |
| db1                |
| dbms18_gl2_friday  |
| dbms2018           |
| dbms2018_friday    |
| dbms2019           |
| lab1               |
| mydb               |
| mysql              |
| performance_schema |
| project            |
| sakila             |
| student            |
| test               |
| world              |
+--------------------+
18 rows in set (0.34 sec)
```

Create Table

```
mysql> create table Employee
    -> (
    -> Emp_ID int(8) AUTO_INCREMENT,
    -> Emp_Name varchar(30) NOT NULL,
    -> Designation varchar(20),
    -> Dept varchar(20),
    -> Salary decimal(8,2),
    -> City varchar (30),
    -> DOB Date,
    -> Marital_Status varchar(15) NOT NULL,
    -> PRIMARY KEY(Emp_ID)
    -> );
Query OK, 0 rows affected (0.17 sec)
```

Display schema of table

```
mysql> describe employee;
+----------------+--------------+------+-----+---------+----------------+
| Field          | Type         | Null | Key | Default | Extra          |
+----------------+--------------+------+-----+---------+----------------+
| Emp_ID         | int(8)       | NO   | PRI | NULL    | auto_increment |
| Emp_Name       | varchar(30)  | NO   |     | NULL    |                |
| Designation    | varchar(20)  | YES  |     | NULL    |                |
| Dept           | varchar(20)  | YES  |     | NULL    |                |
| Salary         | decimal(8,2) | YES  |     | NULL    |                |
| City           | varchar(30)  | YES  |     | NULL    |                |
| DOB            | date         | YES  |     | NULL    |                |
| Marital_Status | varchar(15)  | NO   |     | NULL    |                |
+----------------+--------------+------+-----+---------+----------------+
8 rows in set (0.00 sec)
```

Inserting data

```
mysql> insert into Employee values(1007, 'Alex Bert', 'Manager', 'Finance', 55000, 'Washington', '1979-04-03', 'Married');
Query OK, 1 row affected (0.03 sec)

mysql> insert into Employee values(1008, 'Johnny W', 'Manager', 'Purchase', 50000, 'California', '1980-06-05', 'Married');
Query OK, 1 row affected (0.03 sec)

mysql> insert into Employee values(1009, 'Julia Ronald', 'CFO', 'Finance', 99000.5, 'Alabama', '1981-5-9', 'Unmarried');
Query OK, 1 row affected (0.05 sec)

mysql> insert into Employee values(1010, 'Sandra S', 'Peon', 'Logistics', 20000.4, 'Houston', '1980-12-8', 'Married');
Query OK, 1 row affected (0.05 sec)

mysql> insert into Employee values(1011, 'Cameron L', 'LAM', 'HR', 60000, 'Chicago', '1972-1-12', 'Married');
Query OK, 1 row affected (0.03 sec)

mysql> insert into Employee values(1012, 'Frances Luis', 'Accountant', 'Finance', 61000, 'Washington', '1979-04-03', 'Unmarrie
Query OK, 1 row affected (0.02 sec)

mysql> insert into Employee values(1013, 'Burt Kethy', 'Clerk', 'Purchase', 30000, 'California', '1980-06-05', 'Married');
Query OK, 1 row affected (0.02 sec)

mysql> insert into Employee values(1014, 'Gary Coach', 'Accountant', 'Finance', 40000, 'Alabama', '1981-5-9', 'Unmarried');
Query OK, 1 row affected (0.03 sec)

mysql> insert into Employee values(1015, 'Kevin Luice', 'Driver', 'Logistics', 35000, 'Houston', '1980-12-8', 'Married');
Query OK, 1 row affected (0.02 sec)
```

SELECT QUERIES

(i). Display the table Employee.

```
mysql> select * from employee;
+--------+--------------+-------------+-----------+----------+------------+------------+----------------+
| Emp_ID | Emp_Name     | Designation | Dept      | Salary   | City       | DOB        | Marital_Status |
+--------+--------------+-------------+-----------+----------+------------+------------+----------------+
|   1001 | Reyan Fert   | Manager     | HR        | 80000.00 | California | 1972-01-12 | Married        |
|   1002 | Luther Warn  | Manager     | Finance   | 70000.50 | Washington | 1979-04-03 | Unmarried      |
|   1003 | Audrey Joe   | Associate   | Purchase  | 46000.40 | California | 1980-06-05 | Married        |
|   1004 | Kim John     | Clerk       | Finance   | 30000.00 | Alabama    | 1981-05-09 | Unmarried      |
|   1005 | John Methew  | Driver      | Logistics | 25000.00 | Houston    | 1980-12-08 | Married        |
|   1006 | Mery Lee     | Associate   | HR        | 40000.00 | Chicago    | 1972-01-12 | Unmarried      |
|   1007 | Alex Bert    | Manager     | Finance   | 55000.00 | Washington | 1979-04-03 | Married        |
|   1008 | Johnny W     | Manager     | Purchase  | 50000.00 | California | 1980-06-05 | Married        |
|   1009 | Julia Ronald | CFO         | Finance   | 99000.50 | Alabama    | 1981-05-09 | Unmarried      |
|   1010 | Sandra S     | Peon        | Logistics | 20000.40 | Houston    | 1980-12-08 | Married        |
|   1011 | Cameron L    | LAM         | HR        | 60000.00 | Chicago    | 1972-01-12 | Married        |
|   1012 | Frances Luis | Accountant  | Finance   | 61000.00 | Washington | 1979-04-03 | Unmarried      |
|   1013 | Burt Kethy   | Clerk       | Purchase  | 30000.00 | California | 1980-06-05 | Married        |
|   1014 | Gary Coach   | Accountant  | Finance   | 40000.00 | Alabama    | 1981-05-09 | Unmarried      |
|   1015 | Kevin Luice  | Driver      | Logistics | 35000.00 | Houston    | 1980-12-08 | Married        |
+--------+--------------+-------------+-----------+----------+------------+------------+----------------+
15 rows in set (0.00 sec)
```

(ii). All details of the employees whose Emp_ID=1004.

```
mysql> select * from employee where emp_id=1004;;
+--------+----------+-------------+---------+----------+---------+------------+----------------+
| Emp_ID | Emp_Name | Designation | Dept    | Salary   | City    | DOB        | Marital_Status |
+--------+----------+-------------+---------+----------+---------+------------+----------------+
|   1004 | Kim John | Clerk       | Finance | 30000.00 | Alabama | 1981-05-09 | Unmarried      |
+--------+----------+-------------+---------+----------+---------+------------+----------------+
1 row in set (0.04 sec)
```

(iii). All details of the employees whose **salary >=30000 and <65000.**

```
mysql> select * from employee where salary>=30000 and salary<=65000;
+--------+--------------+-------------+-----------+----------+------------+------------+----------------+
| Emp_ID | Emp_Name     | Designation | Dept      | Salary   | City       | DOB        | Marital_Status |
+--------+--------------+-------------+-----------+----------+------------+------------+----------------+
|   1003 | Audrey Joe   | Associate   | Purchase  | 46000.40 | California | 1980-06-05 | Married        |
|   1004 | Kim John     | Clerk       | Finance   | 30000.00 | Alabama    | 1981-05-09 | Unmarried      |
|   1006 | Mery Lee     | Associate   | HR        | 40000.00 | Chicago    | 1972-01-12 | Unmarried      |
|   1007 | Alex Bert    | Manager     | Finance   | 55000.00 | Washington | 1979-04-03 | Married        |
|   1008 | Johnny W     | Manager     | Purchase  | 50000.00 | California | 1980-06-05 | Married        |
|   1011 | Cameron L    | LAM         | HR        | 60000.00 | Chicago    | 1972-01-12 | Married        |
|   1012 | Frances Luis | Accountant  | Finance   | 61000.00 | Washington | 1979-04-03 | Unmarried      |
|   1013 | Burt Kethy   | Clerk       | Purchase  | 30000.00 | California | 1980-06-05 | Married        |
|   1014 | Gary Coach   | Accountant  | Finance   | 40000.00 | Alabama    | 1981-05-09 | Unmarried      |
|   1015 | Kevin Luice  | Driver      | Logistics | 35000.00 | Houston    | 1980-12-08 | Married        |
+--------+--------------+-------------+-----------+----------+------------+------------+----------------+
10 rows in set (0.00 sec)
```

(iv). All details of employees belong to **HR department** & resides in **Chicago.**

```
mysql> select * from employee where dept='HR' and City='Chicago';
+--------+-----------+-------------+------+----------+---------+------------+----------------+
| Emp_ID | Emp_Name  | Designation | Dept | Salary   | City    | DOB        | Marital_Status |
+--------+-----------+-------------+------+----------+---------+------------+----------------+
|   1006 | Mery Lee  | Associate   | HR   | 40000.00 | Chicago | 1972-01-12 | Unmarried      |
|   1011 | Cameron L | LAM         | HR   | 60000.00 | Chicago | 1972-01-12 | Married        |
+--------+-----------+-------------+------+----------+---------+------------+----------------+
2 rows in set (0.00 sec)
```

(v). All details of those employees, who are married**.**

```
mysql> select * from employee where marital_status='Married';
+--------+------------+-------------+-----------+----------+------------+------------+----------------+
| Emp_ID | Emp_Name   | Designation | Dept      | Salary   | City       | DOB        | Marital_Status |
+--------+------------+-------------+-----------+----------+------------+------------+----------------+
|   1001 | Reyan Fert | Manager     | HR        | 80000.00 | California | 1972-01-12 | Married        |
|   1003 | Audrey Joe | Associate   | Purchase  | 46000.40 | California | 1980-06-05 | Married        |
|   1005 | John Methew | Driver     | Logistics | 25000.00 | Houston    | 1980-12-08 | Married        |
|   1007 | Alex Bert  | Manager     | Finance   | 55000.00 | Washington | 1979-04-03 | Married        |
|   1008 | Johnny W   | Manager     | Purchase  | 50000.00 | California | 1980-06-05 | Married        |
|   1010 | Sandra S   | Peon        | Logistics | 20000.40 | Houston    | 1980-12-08 | Married        |
|   1011 | Cameron L  | LAM         | HR        | 60000.00 | Chicago    | 1972-01-12 | Married        |
|   1013 | Burt Kethy | Clerk       | Purchase  | 30000.00 | California | 1980-06-05 | Married        |
|   1015 | Kevin Luice | Driver     | Logistics | 35000.00 | Houston    | 1980-12-08 | Married        |
+--------+------------+-------------+-----------+----------+------------+------------+----------------+
9 rows in set (0.00 sec)
```

(vi). Display the list of Employee (only names in **ascending & descending** order)

```
mysql> select emp_name from employee order by emp_name asc;
+--------------+
| emp_name     |
+--------------+
| Alex Bert    |
| Audrey Joe   |
| Burt Kethy   |
| Cameron L    |
| Frances Luis |
| Gary Coach   |
| John Methew  |
| Johnny W     |
| Julia Ronald |
| Kevin Luice  |
| Kim John     |
| Luther Warn  |
| Mery Lee     |
| Reyan Fert   |
| Sandra S     |
+--------------+
15 rows in set (0.01 sec)

mysql> select emp_name from employee order by emp_name desc;
+--------------+
| emp_name     |
+--------------+
| Sandra S     |
| Reyan Fert   |
| Mery Lee     |
| Luther Warn  |
| Kim John     |
| Kevin Luice  |
| Julia Ronald |
| Johnny W     |
| John Methew  |
| Gary Coach   |
| Frances Luis |
| Cameron L    |
| Burt Kethy   |
| Audrey Joe   |
| Alex Bert    |
+--------------+
15 rows in set (0.00 sec)

mysql>
```

(vii). How many employees works in Logistics dept.

```
mysql> select count(*) as Log_Emp from employee where dept='Logistics';
+---------+
| Log_Emp |
+---------+
|       3 |
+---------+
1 row in set (0.00 sec)
```

(viii). Number of employees from every city.

```
mysql> select city,count(*) as Employees from employee group by City;
+------------+-----------+
| city       | Employees |
+------------+-----------+
| Alabama    |         3 |
| California |         4 |
| Chicago    |         2 |
| Houston    |         3 |
| Washington |         3 |
+------------+-----------+
5 rows in set (0.00 sec)
```

(ix). Display the list of employee name in Upper & Lower cases.

```
mysql> select upper(emp_name) as Employees from employee ;
+--------------+
| Employees    |
+--------------+
| REYAN FERT   |
| LUTHER WARN  |
| AUDREY JOE   |
| KIM JOHN     |
| JOHN METHEW  |
| MERY LEE     |
| ALEX BERT    |
| JOHNNY W     |
| JULIA RONALD |
| SANDRA S     |
| CAMERON L    |
| FRANCES LUIS |
| BURT KETHY   |
| GARY COACH   |
| KEVIN LUICE  |
+--------------+
15 rows in set (0.00 sec)

mysql> select lower(emp_name) as Employees from employee ;
+--------------+
| Employees    |
+--------------+
| reyan fert   |
| luther warn  |
| audrey joe   |
| kim john     |
| john methew  |
| mery lee     |
| alex bert    |
| johnny w     |
| julia ronald |
| sandra s     |
| cameron l    |
| frances luis |
| burt kethy   |
| gary coach   |
| kevin luice  |
+--------------+
15 rows in set (0.00 sec)
```

(x). Find the details of the employee who are getting **highest, Lowest &** how many employees are getting more than **average** salary**.**

```
mysql> select * from employee where Salary=(select max(Salary) from employee);
+--------+--------------+-------------+---------+----------+---------+------------+----------------+
| Emp_ID | Emp_Name     | Designation | Dept    | Salary   | City    | DOB        | Marital_Status |
+--------+--------------+-------------+---------+----------+---------+------------+----------------+
|   1009 | Julia Ronald | CFO         | Finance | 99000.50 | Alabama | 1981-05-09 | Unmarried      |
+--------+--------------+-------------+---------+----------+---------+------------+----------------+
1 row in set (0.00 sec)

mysql> select * from employee where Salary=(select min(Salary) from employee);
+--------+----------+-------------+-----------+----------+---------+------------+----------------+
| Emp_ID | Emp_Name | Designation | Dept      | Salary   | City    | DOB        | Marital_Status |
+--------+----------+-------------+-----------+----------+---------+------------+----------------+
|   1010 | Sandra S | Peon        | Logistics | 20000.40 | Houston | 1980-12-08 | Married        |
+--------+----------+-------------+-----------+----------+---------+------------+----------------+
1 row in set (0.00 sec)

mysql> select * from employee where Salary>(select avg(Salary) from employee);
+--------+--------------+-------------+----------+----------+------------+------------+----------------+
| Emp_ID | Emp_Name     | Designation | Dept     | Salary   | City       | DOB        | Marital_Status |
+--------+--------------+-------------+----------+----------+------------+------------+----------------+
|   1001 | Reyan Fert   | Manager     | HR       | 80000.00 | California | 1972-01-12 | Married        |
|   1002 | Luther Warn  | Manager     | Finance  | 70000.50 | Washington | 1979-04-03 | Unmarried      |
|   1007 | Alex Bert    | Manager     | Finance  | 55000.00 | Washington | 1979-04-03 | Married        |
|   1008 | Johnny W     | Manager     | Purchase | 50000.00 | California | 1980-06-05 | Married        |
|   1009 | Julia Ronald | CFO         | Finance  | 99000.50 | Alabama    | 1981-05-09 | Unmarried      |
|   1011 | Cameron L    | LAM         | HR       | 60000.00 | Chicago    | 1972-01-12 | Married        |
|   1012 | Frances Luis | Accountant  | Finance  | 61000.00 | Washington | 1979-04-03 | Unmarried      |
+--------+--------------+-------------+----------+----------+------------+------------+----------------+
7 rows in set (0.00 sec)
```

(xi). Display the list of 5 most senior employees**.**

```
mysql> select * from employee
    -> order by DOB
    -> limit 5;
+--------+-------------+-------------+---------+----------+------------+------------+----------------+
| Emp_ID | Emp_Name    | Designation | Dept    | Salary   | City       | DOB        | Marital_Status |
+--------+-------------+-------------+---------+----------+------------+------------+----------------+
|   1001 | Reyan Fert  | Manager     | HR      | 80000.00 | California | 1972-01-12 | Married        |
|   1011 | Cameron L   | LAM         | HR      | 60000.00 | Chicago    | 1972-01-12 | Married        |
|   1006 | Mery Lee    | Associate   | HR      | 40000.00 | Chicago    | 1972-01-12 | Unmarried      |
|   1002 | Luther Warn | Manager     | Finance | 70000.50 | Washington | 1979-04-03 | Unmarried      |
|   1007 | Alex Bert   | Manager     | Finance | 55000.00 | Washington | 1979-04-03 | Married        |
+--------+-------------+-------------+---------+----------+------------+------------+----------------+
5 rows in set (0.00 sec)
```